

Capstone Project Report: Artist Classification

1. Introduction

The "Best Artworks of All Time" project was developed during my studies at Tuwaiq Academy to explore the intersection of machine learning, deep learning, and computer vision. The main goal of this project is to build a classification pipeline capable of identifying the correct artist behind a given painting based entirely on its image pixels. Exploring this dataset allowed me to learn how neural networks and classical machine learning algorithms can analyze visual patterns, brush strokes, textures, and color distributions to distinguish between the unique styles of world-renowned master artists.

2. Dataset Description

The project utilizes the "Best Artworks of All Time" dataset downloaded from Kaggle. While the full dataset tracks biographical metadata for 50 of the world's most famous artists, the image repository actually contains paintings spread across 51 categories.

To optimize model training and ensure a high density of visual samples per class, the data was filtered to focus exclusively on the top 5 most prominent artists. The final training subset consists of the following class distributions:

1. Vincent van Gogh: 877 images
2. Edgar Degas: 702 images
3. Pablo Picasso: 439 images
4. Pierre-Auguste Renoir: 336 images
5. Albrecht Dürer: 328 images

3. Models Used

To establish a baseline for tabular and pixel-based classification, three classical machine learning algorithms were selected from the Scikit-Learn library:

- Logistic Regression: Used to fit linear boundaries between flattened image array features.
- Random Forest Classifier: Used to evaluate non-linear ensemble decision trees over pixel structures.
- K-Nearest Neighbors (KNN): Implemented to group artwork patterns based on spatial feature distances.

Among these classical models, Logistic Regression and Random Forest executed much faster, whereas KNN required substantial memory allocation for distance evaluations over pixel intensities.

4. Neural Network

To capture spatial hierarchies and complex representations within the painting data that traditional machine learning models struggle to interpret, a Multi-Layer Perceptron (MLP) Neural Network was built using the Keras and TensorFlow frameworks.

The network architecture features multiple dense, fully-connected layers (including **Dense(128)** and **Dense(64)** configurations) paired with regularized **Dropout** functions to mitigate overfitting. The categorical outcomes are processed via a final output layer utilizing a softmax activation function to predict probabilities across the 5 target artist classes. The model was trained with the Adam optimizer over 15 epochs.

5. Results & Comparison

After running exhaustive evaluations across all four pipelines, the comparison yields the following observations:

- The Neural Network achieved a higher F1-Score than all 3 classical ML models. * Classical models (such as Logistic Regression and Random Forest) trained much quicker but struggled to optimize directly on raw, complex pixel formats without extensive feature engineering.
- The Deep Learning approach successfully extracted abstract stylistic patterns from the pixel matrices. Despite requiring a longer training duration, the significant jump in classification precision and F1-score makes the computational overhead entirely worthwhile for an image processing task of this scale.

6. What I Learned

Through this capstone project at Tuwaiq Academy, I gained valuable, hands-on experience in computer vision pipelines. I learned that:

1. Proper data cleaning and targeting structural thresholds (like narrowing down to the top 5 artists with higher image counts) yields cleaner boundaries and less noisy gradients.
2. For image datasets, flat representations severely limit classical models, highlighting why deep learning structures are crucial for modern visual artificial intelligence tasks.
3. Balancing model complexity against training execution times is critical when designing reliable artificial intelligence workflows.

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